Om Deshmukh

Software Engineer | C++17 / Python | Linux Systems | Trading Infrastructure
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EDUCATION

JSPM University — Pune, India
M.Tech, Computer Science. CGPA: 8.0 / 10 — 2025 - 2027 Expected
AISSMS College of Engineering — Pune, India
B.E., Electrical Engineering. CGPA: 7.5 / 10 — 2019 - 2023

TECHNICAL SKILLS

Languages: C++17, Python, Java, SQL
Linux / Systems: Linux/WSL, userland/kernel fundamentals, processes, threads, memory management, file I/O, TCP/UDP, sockets, debugging, profiling, CMake, Git, Docker
Trading / Quant: STL, price-time priority, order book design, matching logic, integer tick pricing, replay testing, latency benchmarking, Monte Carlo simulation, VaR/Expected Shortfall
Data / Storage: PostgreSQL, Redis, composite indexing, caching, data validation, normalization, reconciliation pipelines, GitHub Actions, AWS EC2/S3

WORK EXPERIENCE

CapSurge — Pune, India
Software Engineer (Full-time) | Financial Data Automation — Mar 2026 - Present

- Building an automation engine for financial modeling workflows that ingests provider data, validates records, normalizes financial inputs, and produces reproducible reconciliation outputs for analyst review.
- Built tenant-isolated financial data services with composite PostgreSQL indexing, processing 2,200+ records across 13 providers monthly while preserving strict data boundaries.
- Implemented an idempotent reconciliation pipeline with deduplication keys, validation checks, and structured exception handling for financial-data workflows.
- Reduced integration turnaround by approximately 40% by defining structured data contracts and reproducible workflow outputs for analyst-facing finance operations.

CodeClause Pvt Ltd — Remote
Software Engineering Intern | PostgreSQL / Redis / CI — Oct 2023 - Nov 2023

- Reduced endpoint latency by 20% using PostgreSQL indexes, optimized joins, and Redis caching; automated CI workflow with GitHub Actions.

PROJECTS

C++ Limit Order Book & Matching Engine — C++17, STL, CMake

- Built a deterministic order book supporting bid/ask levels, price-time priority, partial fills, cancellation, and top-of-book snapshots using integer tick pricing.
- Designed order lifecycle and correctness checks for FIFO matching, crossing logic, cancel lookup, book-state validation, and complexity analysis.
- Extending the engine with replay-driven tests, p50/p95/p99 latency benchmarks, monitoring-style logs, and Linux profiling for trading infrastructure evaluation.

Linux Systems Trading Lab — In Progress | C++, Linux/WSL, Sockets

- Implementing focused experiments for process/thread behavior, TCP socket messaging, file I/O buffering, memory-layout comparison, and timing with std::chrono.
- Building a reusable benchmark harness to compare latency distributions and regression behavior across trading-system components.

Monte Carlo Risk Simulation Lab — In Progress | Python, NumPy, Risk Metrics

- Building GBM-based simulation experiments for terminal price distributions, 95% Value at Risk, Expected Shortfall, and stress-style risk scenarios.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- LeetCode Knight - solved 350 DSA problems across graphs, dynamic programming, trees, greedy algorithms, binary search, and implementation-heavy contests.
- University-level winner, Smart India Hackathon 2026; cleared Google SWE Apprenticeship Technical Round 1 in 2024.